

iClusterBayes Summary Report

Automated Report

Hyperparameters

```
cat('sdev = 0.03, beta.var.scale = 0.2\n')
```

```
## sdev = 0.03, beta.var.scale = 0.2
```

Key Summary Statistics

```
cat(paste0('```\n', '
--- Summary for K = 1 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
  0.3572  0.3794  0.3872  0.3876  0.3961  0.4200

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.09444 0.13833 0.13432 0.16889 0.86500
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1         1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.09667 0.14111 0.13698 0.17222 0.80889
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.005000 0.008333 0.102236 0.031667 1.000000
```

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Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.001111 0.005000 0.008333 0.077411 0.024444 1.000000
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0005556 0.0061111 0.0150000 0.1625307 0.1188889 1.0000000

--- Summary for K = 2 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
  0.1689  0.2111  0.2183  0.2185  0.2261  0.2517

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.04722 0.09000 0.09689 0.13000 0.93111
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.05000 0.09333 0.09915 0.13333 0.67500
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.001111 0.036111 0.401883 1.000000 1.000000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000000 0.0005556 0.0027778 0.2367719 0.2611111 1.0000000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.07889 1.00000 0.68540 1.00000 1.00000

--- Summary for K = 3 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.03222 0.04722 0.05222 0.05264 0.05722 0.08278

```

```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.02889 0.07417 0.07866 0.11167 0.81667
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.03000 0.07833 0.08209 0.11667 0.62333
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000  0.9600  1.0000  0.8024  1.0000  1.0000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.000000 0.001111 0.229487 0.209444 1.000000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000  0.0900  1.0000  0.6855  1.0000  1.0000

--- Summary for K = 4 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.01278 0.02111 0.02500 0.02538 0.02944 0.06222

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.02278 0.06139 0.06889 0.09722 0.81944
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 5 :

```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02444	0.06556	0.07283	0.10444	0.57889

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.7506	1.0000	0.7752	1.0000	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2713	0.6944	1.0000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000	1.000	1.000	0.862	1.000	1.000

--- Summary for K = 5 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.002778	0.010000	0.013333	0.014263	0.017778	0.068333

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02000	0.05333	0.06204	0.08778	0.79833

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02222	0.05778	0.06660	0.09500	0.58556

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.7333	1.0000	0.7873	1.0000	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
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    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 0.0000  0.0000  0.2911  0.8629  1.0000
Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 1.0000  1.0000  0.8973  1.0000  1.0000

--- Summary for K = 6 ---
Z.ar:
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.003333 0.013333 0.016667 0.017638 0.020556 0.074444

beta.ar:
Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.01056 0.04000 0.05169 0.07556 0.78167
Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1         1         1         1         1         1
Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1         1         1         1         1         1
Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1         1         1         1         1         1
Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1         1         1         1         1         1

gamma.ar:
Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.01222 0.04444 0.05573 0.08333 0.55111
Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.07444 0.68500 0.57611 1.00000 1.00000
Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.8578  1.0000  1.0000  1.0000  1.0000  1.0000
Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000  0.0000  0.0000  0.2694  0.6467  1.0000
Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000  0.8933  1.0000  0.8430  1.0000  1.0000

--- Summary for K = 7 ---
Z.ar:
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.001667 0.007778 0.010000 0.010877 0.012778 0.055000

beta.ar:
Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.

```

0.000000 0.006111 0.028333 0.040526 0.060000 0.749444

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.006667	0.032222	0.044602	0.067778	0.531111

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.7944	0.5892	1.0000	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.4206	1.0000	1.0000	0.9999	1.0000	1.0000

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2063	0.0325	1.0000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.4283	0.9556	0.7035	1.0000	1.0000

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.01556	0.02722	0.03389	0.03544	0.04111	0.18000

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.002778	0.021111	0.034941	0.053333	0.733333

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

```
gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.003333 0.024444 0.039098 0.061111 0.501667
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.0000 0.0000 0.4917 0.4431 0.8128 1.0000
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.1217 0.9333 0.9828 0.9570 1.0000 1.0000
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.00000 0.00000 0.00000 0.09484 0.00000 1.00000
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.0000 0.0000 0.3461 0.3914 0.7456 1.0000
```

--- Summary for K = 9 ---

```
Z.ar:
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.01889 0.03444 0.03944 0.04119 0.04667 0.12111
```

```
beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.001667 0.013333 0.028533 0.042222 0.688333
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
1 1 1 1 1 1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
1 1 1 1 1 1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
1 1 1 1 1 1
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
1 1 1 1 1 1
```

```
gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.002222 0.015556 0.032147 0.048889 0.475556
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.0000 0.0000 0.0000 0.2705 0.5844 1.0000
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.0000 0.8400 0.9306 0.8972 1.0000 1.0000
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.00000 0.00000 0.00000 0.06796 0.00000 1.00000
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
```

```
0.00000 0.00000 0.01222 0.29155 0.59833 1.00000
', '\n`'))
```

```
## ' '
##
##
## --- Summary for K = 1 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.3572 0.3794 0.3872 0.3876 0.3961 0.4200
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.09444 0.13833 0.13432 0.16889 0.86500
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.09667 0.14111 0.13698 0.17222 0.80889
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.005000 0.008333 0.102236 0.031667 1.000000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.001111 0.005000 0.008333 0.077411 0.024444 1.000000
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0005556 0.0061111 0.0150000 0.1625307 0.1188889 1.0000000
##
## --- Summary for K = 2 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.1689 0.2111 0.2183 0.2185 0.2261 0.2517
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
```



```

## 0.00000 0.04722 0.09000 0.09689 0.13000 0.93111
## Modality 2 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 3 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 4 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 5 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
##
## gamma.ar:
## Modality 1 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.00000 0.05000 0.09333 0.09915 0.13333 0.67500
## Modality 2 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.000000 0.001111 0.036111 0.401883 1.000000 1.000000
## Modality 3 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 4 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.0000000 0.0005556 0.0027778 0.2367719 0.2611111 1.0000000
## Modality 5 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.00000 0.07889 1.00000 0.68540 1.00000 1.00000
##
## --- Summary for K = 3 ---
## Z.ar:
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.03222 0.04722 0.05222 0.05264 0.05722 0.08278
##
## beta.ar:
## Modality 1 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.00000 0.02889 0.07417 0.07866 0.11167 0.81667
## Modality 2 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 3 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 4 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 5 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
##
## gamma.ar:

```

```

## Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.03000 0.07833 0.08209 0.11667 0.62333
## Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.9600  1.0000  0.8024  1.0000  1.0000
## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.000000 0.001111 0.229487 0.209444 1.000000
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0900  1.0000  0.6855  1.0000  1.0000
##
## --- Summary for K = 4 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.01278 0.02111 0.02500 0.02538 0.02944 0.06222
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.02278 0.06139 0.06889 0.09722 0.81944
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.02444 0.06556 0.07283 0.10444 0.57889
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.7506  1.0000  0.7752  1.0000  1.0000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.0000  0.2713  0.6944  1.0000
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000  1.000  1.000  0.862  1.000  1.000

```

```

##
## --- Summary for K = 5 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.002778 0.010000 0.013333 0.014263 0.017778 0.068333
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.02000 0.05333 0.06204 0.08778 0.79833
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.02222 0.05778 0.06660 0.09500 0.58556
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.7333  1.0000  0.7873  1.0000  1.0000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.0000  0.2911  0.8629  1.0000
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  1.0000  1.0000  0.8973  1.0000  1.0000
##
## --- Summary for K = 6 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.003333 0.013333 0.016667 0.017638 0.020556 0.074444
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.01056 0.04000 0.05169 0.07556 0.78167
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.

```

```

##      1      1      1      1      1      1
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.01222 0.04444 0.05573 0.08333 0.55111
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.07444 0.68500 0.57611 1.00000 1.00000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.8578  1.0000  1.0000  1.0000  1.0000  1.0000
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.0000  0.2694  0.6467  1.0000
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.8933  1.0000  0.8430  1.0000  1.0000
##
## --- Summary for K = 7 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.001667 0.007778 0.010000 0.010877 0.012778 0.055000
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.006111 0.028333 0.040526 0.060000 0.749444
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.006667 0.032222 0.044602 0.067778 0.531111
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.7944  0.5892  1.0000  1.0000

```

```

## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.4206  1.0000  1.0000  0.9999  1.0000  1.0000
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.0000  0.2063  0.0325  1.0000
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.4283  0.9556  0.7035  1.0000  1.0000
##
## --- Summary for K = 8 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.01556 0.02722 0.03389 0.03544 0.04111 0.18000
##
## beta.ar:
## Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.002778 0.021111 0.034941 0.053333 0.733333
## Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
## Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.003333 0.024444 0.039098 0.061111 0.501667
## Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.4917  0.4431  0.8128  1.0000
## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.1217  0.9333  0.9828  0.9570  1.0000  1.0000
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.00000 0.00000 0.09484 0.00000 1.00000
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.3461  0.3914  0.7456  1.0000
##
## --- Summary for K = 9 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.01889 0.03444 0.03944 0.04119 0.04667 0.12111
##

```

```

## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.000000 0.001667 0.013333 0.028533 0.042222 0.688333
##   Modality 2 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.000000 0.002222 0.015556 0.032147 0.048889 0.475556
##   Modality 2 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.0000 0.0000 0.0000 0.2705 0.5844 1.0000
##   Modality 3 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.0000 0.8400 0.9306 0.8972 1.0000 1.0000
##   Modality 4 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.00000 0.00000 0.00000 0.06796 0.00000 1.00000
##   Modality 5 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.00000 0.00000 0.01222 0.29155 0.59833 1.00000
##
## ‘‘‘

```

Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1–9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.